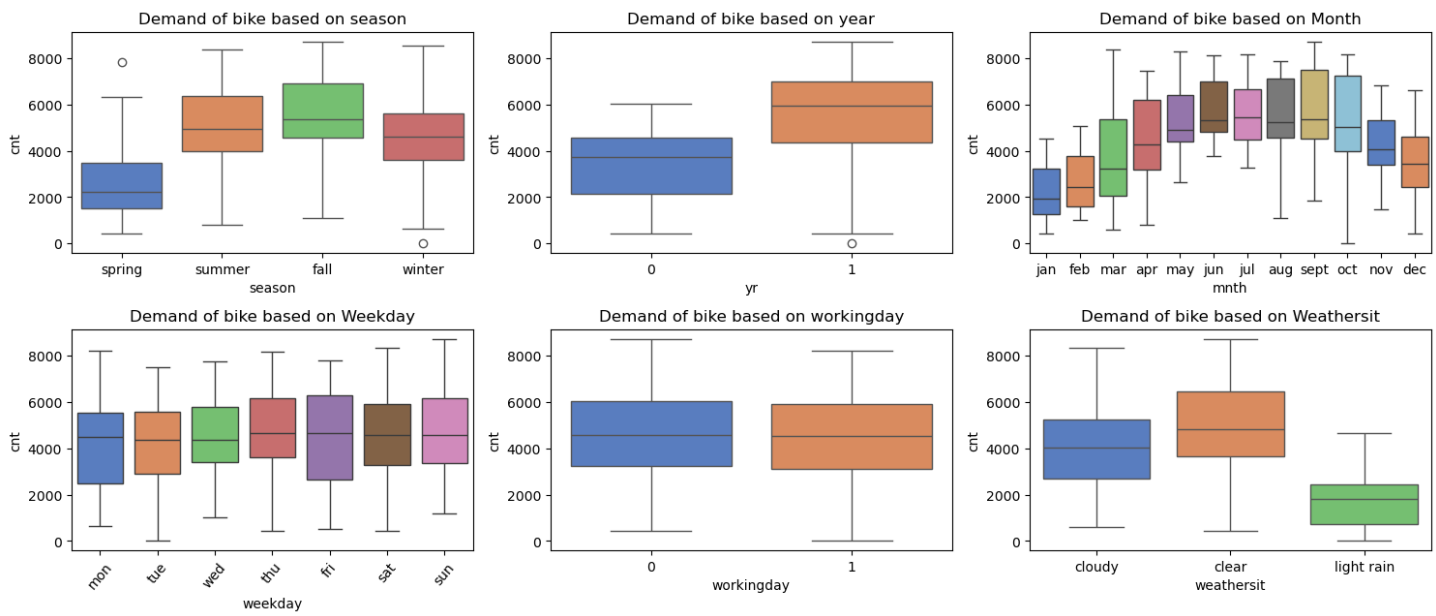
Assignment-based Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

ANS- As ‘cnt’ has considered as a target variable which we consider it as a dependent variable. Categorical variable such as ‘season’, ‘yr’, ’month’ , ‘weekdays’, ’workingdays’ and ‘weathersit’ has their effect on Dependent variable.



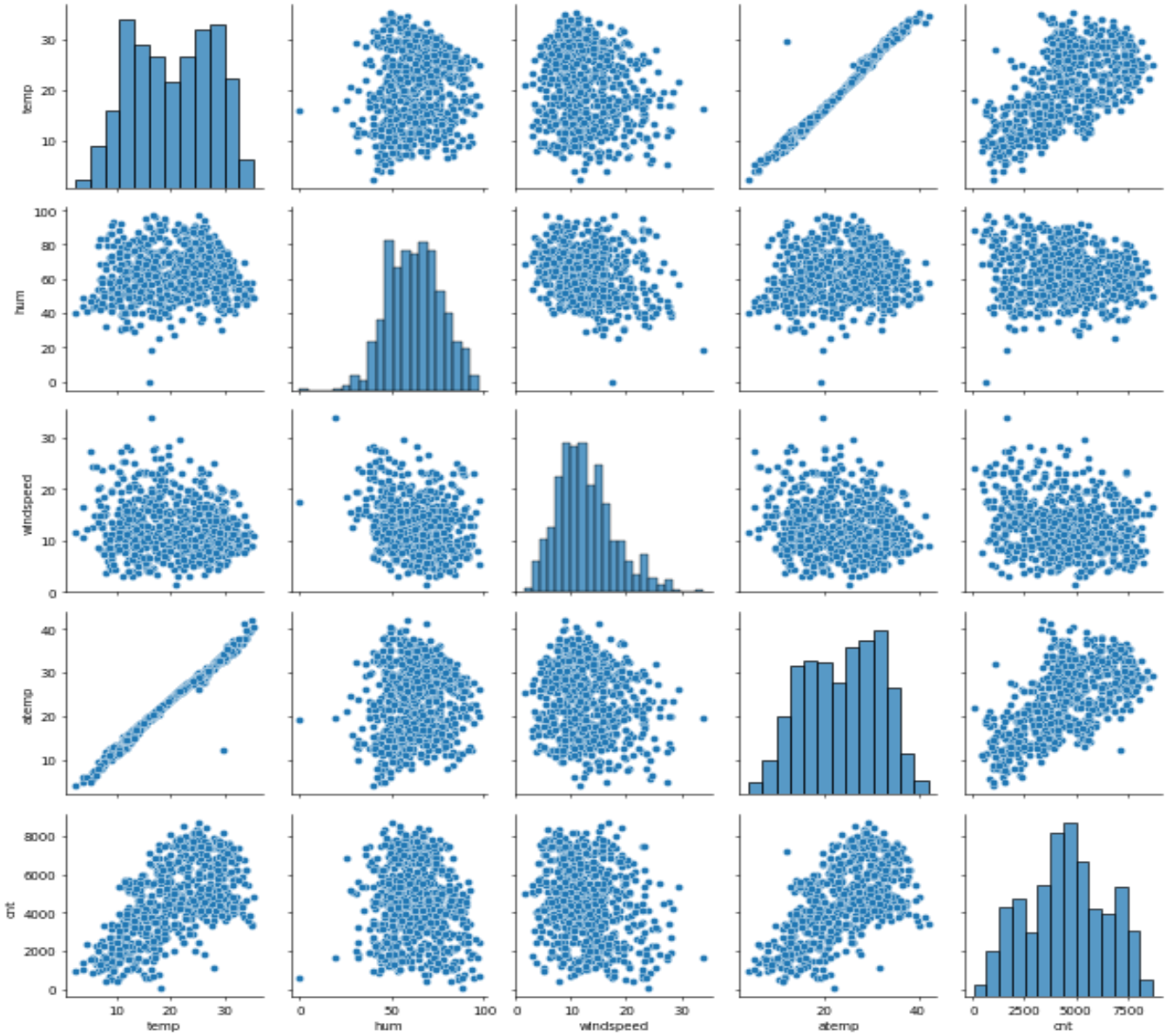
2. Why is it important to use drop_first=True during dummy variable creation?

ANS- drop_first=True will drop the one of the dummy columns that we have created. It important to use drop_first=True during dummy variable creation as for the analysis purpose of dummy variable is to used to represent categorical data in a numerical format. For a categorical variable of “n” levels we will create a “n-1” new columns . Its absence or presence is indicated in terms of 0 and 1 respectively.

For Example- If there are 5 levels by using **drop_first=True** the first columns get eliminated. So we will get 4 columns.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

ANS- While comparing with the target variable ('cnt'), the 'temp' and 'atemp' variables are highly correlated. As it could be shown in the image given below.



4. How did you validate the assumptions of Linear Regression after building the model on the training set?

ANS- While building a Linear Regression there are certain assumption that needed to be followed to make the result reliable.

1. Linerarity – To make the variables linear there must be relationship between the response and predictor.

2. Independence of Errors – It will lead to misleading result, if autocorrelation exists.

3. No Multicollinerarity- if multicollinerarity exists we will need to delect/ remove the correlated variables.

4. Normality of Residual

5. Balance between Observation and Predictor- If there is no balnce it will lead to overfitting, the model will become over complex.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

ANS- Based on the final model, the top 3 features contributing significantly towards explaining the demand of the shared bikes are 'atemp' , 'weathersit_lightrain' and 'yr'

General Subjective Questions

1. Explain the linear regression algorithm in detail.

ANS- Regression means showing relationship between the variables and making the prediction accordingly i.e relationship between dependent and independent variables. The equation of linear regression

I. *simple linear regression* is **Y=mX +b**

Where

- Y is dependent variable
- X id independent variable
- m is the slope of line
- b is intercept

Linear Regression is an statistical method.

Two Factors to be examine for Linear Regression are:

1. Which independent variable is more significant for dependent variable.

2. It is important to have best fit line for highest possible accuracy

II. Multiple Linear Regression:

$Y=\beta 0+\beta 1X1+\beta 2X2+\ldots+\beta pXp+\epsilon$

The equation of the best fit regression line $Y = \beta_o + \beta_1 X$ can be found by the following two methods:

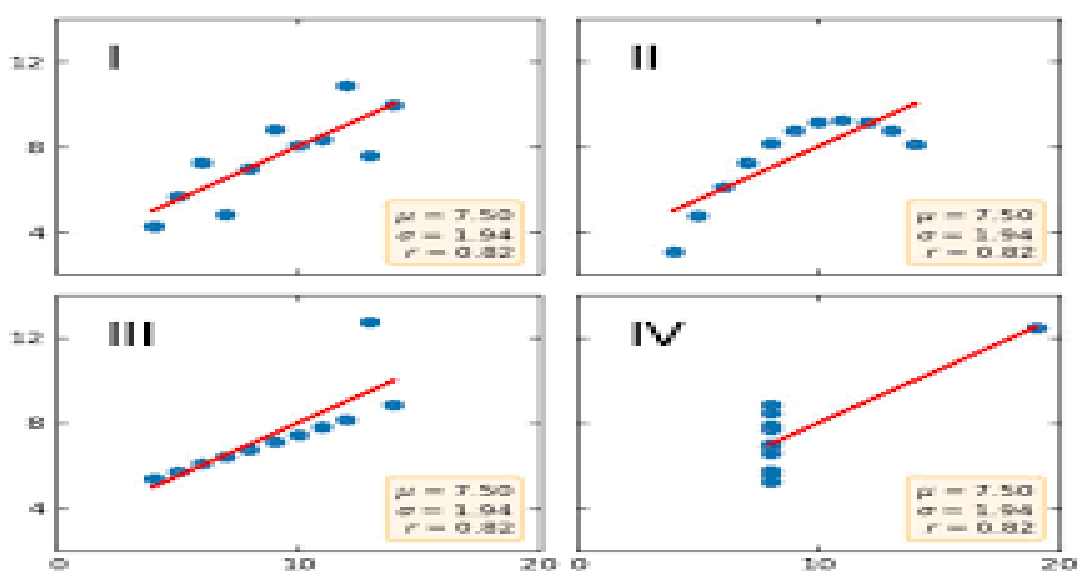
A)Differentiation

B)Gradient descent

T**o find the best fit line-** Minimizing the Distance between the line and the data points like sum of squared errors, Sum of Absolute Errors , Root Mean Squared etc. Linear Regression is about drawing a best fitting line over a data. The core idea is to draw a line that fits the data in best way

2. Explain the Anscombe's quartet in detail.

ANS- Anscombe's Quartet is the model example to demonstrate the importance of data to signify both the importance of plotting data before analyzing it with statistical properties. It contains the set of four data-set and each data-set consists of eleven (X,Y) points. They all share the identical descriptive statistics(mean, variance, standard deviation etc) but different graphical representation. When plotted on scatter plot they appear different. Each graph plot shows the different behavior despite of statistical analysis. These are the 4 data –set



ABOUT MODEL-

I. 1st-As there is a linear relationship between X and Y so it fits linear regression model

II. 2nd-As there is a no linear relationship between Xand Y so it does not fits linear regression model

III. 3rd- There are some outliers that linear regression model does not handle.

IV. 4th-high liverage point sit indicates high correlation coefficient.

3. What is Pearson's R?

ANS- Pearson Correlation Coefficient (PCC) also known statistical term which is also known Pearson's R. It means the correlation between the two set of data. It ranges between -1 to 1

Pearson Correlation Coefficient(r)	Correlation	Interpretation
Between 0 and 1	Positive correlation	Same direction
0	No correlation	No relationship
Between -1 and 0	Negative Correlation	Opposite direction

Formula

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\, n\Sigma x^2 - (\Sigma x)^2 \,]\,[\, n\Sigma y^2 - (\Sigma y)^2 \,]}}$$

r= correlation coefficient

4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

ANS- Scaling is the process of transforming the values of the dataset within the specific range. Range could be either 0-1 or 0-100. It ensures that all the features contribute equally.Scaling the variable is important as it help different variable to same line or we can sayon equal foot.

Example- Considering the price . It could have values in U.S dollars and in Indian Ruppes . As we know the value of 1 USD = 85.4827 INR . Due to this we can get Error while measuring SVM and KNN. That is the reason we need to scale

The difference between normalized scaling and standardized scaling is that , the scaling we change the range of the data while on the other hand in normalization, we change the shape of distribution of your data. Normalization is the more radical transformation.

- Standardised scaling is not affected by outliers where as normalized scaling does.
- Standardised scaling is used when distribution is normal where as normalized scaling is used when we don't know about distribution.
- Standardised scaling is called as Z-Score Normalization whereas Normalized scaling is called as scaling normalization.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

ANS- When there is perfect multicollinerarity VIF becomes infinite in a regression model. It basically indicate that Higher the VIF , the higher the degree of multicollineraity. VIF helps in explaining the relationship with one independent variable with all the other independent variable. VIF greater than 10 it is definitely high. VIF greater than 5 must not be ignored and it must be removed/drop.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

ANS- Two distributions' quantiles (or percentiles) can be compared using a Q-Q plot. One distribution is typically the observed data and the other is a theoretical or reference distribution (the normal distribution).

If the data is in line with the theoretical distribution, the Q-Q plot points will be roughly on a straight line. The data's lack of conformity to the theoretical distribution can be conclude from the deviations from a straight line.

Use and importance of a Q-Q plot in linear regression

1. Q-Q plots can assist in identifying outliers, which are data points that differ significantly from the overall pattern.
2. By utilizing Q-Q plots, you can visually assess the normality of the residuals, making it easier to identify potential issues.
3. Q-Q plots is used with sample size as well.
4. While creating train-test dataset we create Q-Q plots.